

ESAC: An Algorithm for Fissure Line Detection

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Abstract—Pulmonary fissure detection is an important step for lung lobe segmentation which is necessary for accurate diagnostics and surgical planning. Automatic detection of fissures in CT images is a challenging task due to varying intensity, pathological deformation and noisy acquisitions. In this paper, we propose a novel fissure line detection technique using eigen analysis of the hessian matrix and an exhaustive sample consensus (ESAC) based line fitting in small overlapping windows. The idea behind using the line fitting technique is that the fissure line appears as piece-wise linear segment in a small window. As opposed to RANSAC, the point selection mechanism in the proposed method chooses all combination of data points exhaustively. This approach reduces the possibility of missing the possible candidate points for a fissure line. Our main contribution lies in detection of the fissure line without using any training data as well as any template matching model. The performance of our method is validated on the publicly available LOLA11 database. Comparisons with some existing approaches on this database indicate the advantage of the proposed solution.

Index Terms—Hessian Matrix, Eigen Analysis, Fissure Line, Sample Consensus

I. INTRODUCTION

A human lung is divided into five lobes where the right lung consists of three lobes and the left lung consists of two lobes [1] and the inter-lobar border is defined as pulmonary fissure. Accurate detection of the fissures is of clinical importance in localization of lesions and in the assessment of lung disease on a lobar level [2]. For detection of pulmonary disease, an X-ray based imaging method known as Thoracic Computed Tomography (CT) is performed. The segmentation of lung lobes in CT images has important applications in disease assessment and surgical planning. However, automated or computer-aided segmentation of pulmonary fissures in CT images is a challenging task. Some of the challenges are varying shape, homogeneous intensity, pathological deformation, similar to vessel like structures, bronchi and fibrotic tissues along with imaging noise.

In this paper, our main purpose is to detect the fissure lines in an unsupervised manner. Due to varying nature of the fissure line we cannot rely on any prior information unlike template or model matching methods. This motivated us in designing a line detection algorithm based on the concept of random sample consensus (RANSAC) [3] model. As opposed to RANSAC, the point selection mechanism in our method differs in a way that it chooses all combination of data points exhaustively

rather selecting it randomly. This approach is more appropriate as the number of points over the fissure line are quite less among all data points. So, it reduces the possibility of missing the set of points that better lie on the fissure line. Our line detection mechanism is termed as exhaustive sample consensus (ESAC). It seeks a linear structure that runs from one end to another of the lung boundary. In our model, we detect the linear structures on a set of overlapping windows over the image. We have utilized the fact that the fissure line appears as a piece-wise linear segment in a small window. A cost function is designed for the selection of the set of data points that best resemble the fissure line if a line is drawn through them. For final extraction, the line for which endpoints either touch the boundary or is very close to the boundary is selected and detected as desired fissure line.

The rest of the paper is organized as follows: In section II, we discuss the related work and highlight our contributions. In section III, we discuss our proposed model of fissure line detection. In section IV, we describe our experimental results with detailed analysis. Finally, the paper is concluded in section V with an outline direction for future research.

II. RELATED WORK

Several automatic and semi-automatic methods have been reported in literature to realize pulmonary fissure segmentation. Saita et al. [4] proposed a fissure extraction algorithm by identifying the region containing fissure based on 3D distance from the lung blood vessel. Kubo et al. [5] presented a 2D VanderBrug operator to enhance the linear structures and simultaneously suppress streak artifacts. Appia et al. [6] proposed a semi-automated minimal path approach to segment fissures on CT images. Wei et al. [7] employed a method for fissure recognition using texture analysis, fissure region analysis and fissure identification. Rixroot et al. [8] presented a multiphase supervised approach to enhance the fissure like structures in 3D chest CT scans. Pu et al. [9] developed an automated scheme using computational geometry to detect and segment fissures in CT scans. Pulmonary fissures are enhanced using iterative laplacian smoothing and approximated using a set of plane patches. Zhang et al. [10] employed a 2D ridgeness measure to enhance fissure contrast. The direction and intensity continuity were integrated to extract the fissure objects using a fuzzy reasoning framework. Wang et al. [11] used a

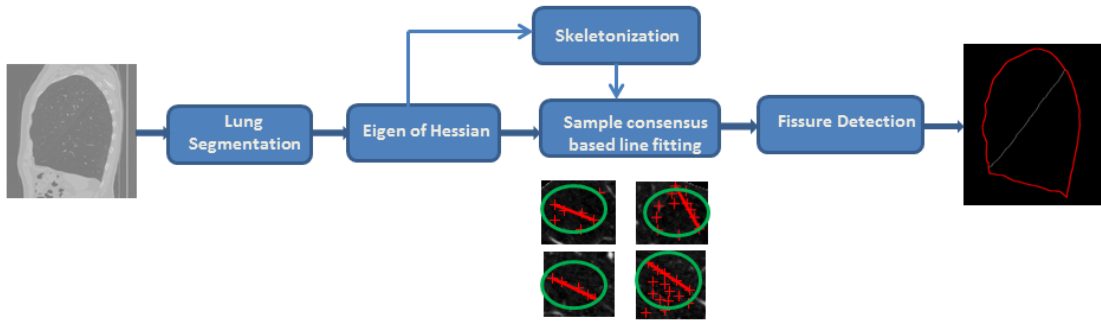


Fig. 1: Different stages of our proposed pipeline.

prior knowledge of the shape of the fissure and constructed a ridge map to enhance the appearance of the fissure on CT scans. Klinder et al. [12] presented a line enhancing filter using multiple hypothesis testing. A template matching between a predefined fissure model and the image content have been performed for line-like structures in two-dimensional cross sections. Xiao et al. [13] proposed a derivative of stick filter for the enhancement of fissure lines in thoracic CT scans. The filter was used in varying directions along a triple stick kernel by defining non-linear derivatives. A post-processing step for fissure segmentation was employed by using multi threshold and branch point removal algorithm. Bragman et al.[14] proposed a lobe segmentation algorithm based on unsupervised fissure segmentation with the creation of group-wise fissure prior and a cost function combining patient and population information.

Most of the existing techniques for fissure line detection in CT images are based on learning approach and template matching. Unlike above mentioned methods, we have proposed a novel unsupervised fissure detection model based on eigen analysis of Hessian filter and exhaustive sample consensus (ESAC) based line detection technique.

III. PROPOSED FRAMEWORK

This section describes our overall methodology for detecting fissure region in thoracic CT images. Presently, our methods is limited to 2D-line extraction in each of the sagittal, coronal and axial cross sectional cuts. Our proposed pipeline consists of three major steps: (1) lung field segmentation (2) eigen analysis of Hessian matrix (3) line fitting based on the proposed exhaustive sample consensus (ESAC) algorithm and finally extracting fissure line.

A. Lung Field Segmentation

The thoracic CT image is first smoothed using a Gaussian filter to reduce noise. A dual threshold is applied on the CT image to extract the lung region. Finally, a morphological hole filling and morphological closing operation is performed to accurately segment the lung field from the CT image.

B. Eigen Analysis of Hessian Matrix

Let the segmented lung field be represented as $L(x, y)$, where, (x, y) denote the corresponding coordinates of the pixel

location. After segmentation, we perform an eigen analysis based on Hessian matrix [15] to extract the local curvature information belonging to fissure lines. The Hessian matrix $H(x, y)$ built at each pixel of the lung field images $L(x, y)$ is composed of four independent second order derivatives as:

$$H(x, y) = \begin{bmatrix} D_{xx}(x, y) & D_{xy}(x, y) \\ D_{yx}(x, y) & D_{yy}(x, y) \end{bmatrix}$$

Let, the eigen values computed at each pixel from the above Hessian matrix are denoted by λ_1 and λ_2 . From the basic definition of eigen analysis, the eigen values must satisfy the criteria $\lambda_1(x, y) < 0$ and $\lambda_2(x, y) < 0$ for a local maxima to occur at a pixel position (x, y) . Thus by setting this criteria, we extract some discrete regions corresponding to a local maxima. After extracting the regions, we assemble pixels belonging to the maxima using connected component labelling. We also compute the centroids as $c(x, y)$ for each of these connected component regions. These centroids will act as a basis for line fitting in our proposed exhaustive sample consensus (ESAC) algorithm. In addition to centroid extraction, we also generate a binary map $S(x, y)$ separately over $H(x, y)$ by setting the eigen value $\lambda_1(x, y) < 0$ followed by skeletonization. This skeleton map preserves the overall structure of fissure line along with all the centroids lying inside $S(x, y)$ (i.e. $c(x, y) \subset S(x, y)$). The extracted path of the fissure is based on the paths of the skeleton map $S(x, y)$.

C. Line Fitting Using Exhaustive Sample Consensus

A fissure line appears as a thin curvilinear structure in 2D slices which is quite difficult to detect especially when the lung regions are mostly homogeneous. To circumvent this problem, we constructed a piece-wise line detection approach in a certain set of overlapped windows w each having a dimension $N \times N$ with $N = 50$ and an overlap ratio of 0.85. The main idea behind such approach is to view the fissure line structure as more distinctive and more linear in a local region. Now, we have tried to comprehend the best fit line within this window based on the centroid set C belonging to this window. Suppose, there are n set of centroids in each window. We exhaustively select $\binom{n}{2}$ combination of the centroids. For each such combinations we fit a line passing through these points. For successful fitting of the line we keep those set

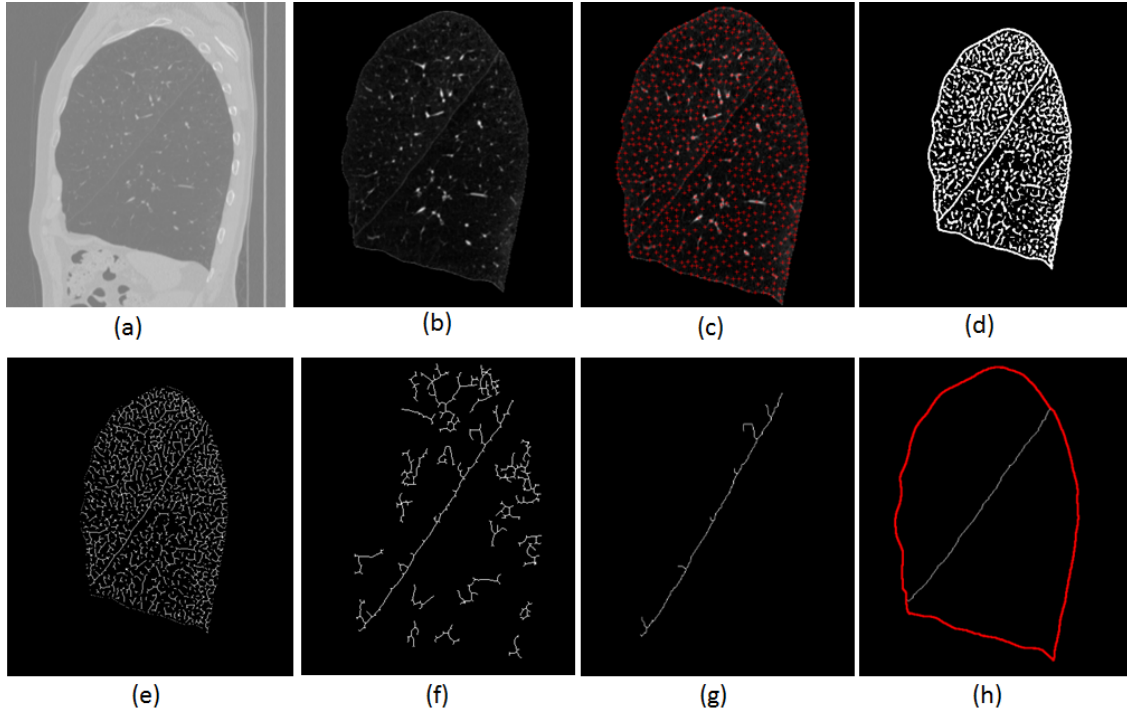


Fig. 2: Output at different stages of our proposed pipeline, where, (a) Chest CT scan; (b) Segmented lung field; (c) Centroid points for $\lambda_1 < 0$ and $\lambda_2 < 0$; (d) $\lambda_1 < 0$; (e) skeleton map from $\lambda_1 < 0$; (f) skeleton map after area threshold; (g) line detection result; (h) final detected fissure line.

of combinations $\#C \leq \binom{n}{2}$ (where $\#$ denotes cardinality of a set) for line fitting if there are sufficient number of

Algorithm 1 Exhaustive sample consensus based fissure line detection

Require: Chest CT scan

Ensure: $f_{line}(x, y)$

- 1: **for** $t = 1$ **to** T **do** /* T is the total no. of slices*/
 - 2: Obtain Lung field $L(x, y)$ using dual threshold and morphological closing
 - 3: Obtain Eigen values λ_1 and λ_2 of the Hessian matrix $H(x, y)$
 - 4: Obtain centroids by setting $\lambda_1 < 0$ and $\lambda_2 < 0$
 - 5: Obtain skeleton image $S(x, y)$ by setting $\lambda_1 < 0$
 - 6: Obtain $\binom{n}{2}$ combination of centroids C over the overlapped window w of size 50×50 and it is repeated on each sub-image
 - 7: Filter centroid combinations based on the minimum inlier of 2 based on equation 1
 - 8: Obtain the best fit line based on the cost function 3
 - 9: Extract fissure line based on equation 4 in each window w
 - 10: Combine the extracted result to get fissure line image $F_{line}(x, y)$
 - 11: Finally detect line with end points closer to lung boundary as $f_{line}(x, y)$
 - 12: **end for**
-

centroids as inliers within the desired line. Also, we want to restrict the number of centroids outside the desired line. So, we bounded the inliers based on a distance threshold (ρ_{th}) from the fitted line. Let, a line be fitted between the centroids $\{c_{i1}(x_1, y_1), c_{i2}(x_2, y_2)\} \in C$. The distance computed for a centroid point $c_j(x_0, y_0)$ is given as:

$$\rho_0 = \frac{|(y_2 - y_1)x_0 + (x_2 - x_1)y_0 + x_2y_1 - y_2x_1|}{\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}} \quad (1)$$

where, $j \neq i1, i2$. Thus if $|\rho_0| < |\rho_{th}|$ then the centroid $c_j(x_0, y_0)$ is declared as an inlier. For fitting a line between any pair of centroids c_{i1}, c_{i2} the total number of inliers must be greater than 2. The number of inliers increases with ρ_0 which increases the computation time but the performance saturates at a certain point (ρ_{th}). Thus, the ρ_{th} is experimentally selected as a typical lowest point where the AUC saturates for most of the CT scan images. In our method, we set ρ_{th} to 4.5. Please see Fig.5 in this context. Now, there may exist more than one path between c_{i1} and c_{i2} but the one with shortest path is assumed to be a more appropriate fit for a fissure line in the selected window w . Let $P_{i1,i2}(x, y) \in S(x, y)$ be the path connecting c_{i1} and c_{i2} with a path length of $l_{i1,i2}$. The shortest path between the centroids c_{i1} and c_{i2} in the window w is obtained using

$$SP_{w,i1,i2}(x, y) = \{P_{i1,i2}(x, y) : \min_{l_{i1,i2}} P_{i1,i2}(x, y)\} \quad (2)$$

Next, the mean of the intensity $\mu_{i1,i2}$ and variance of the intensity $\sigma_{i1,i2}$ of the pixels lying on the shortest path

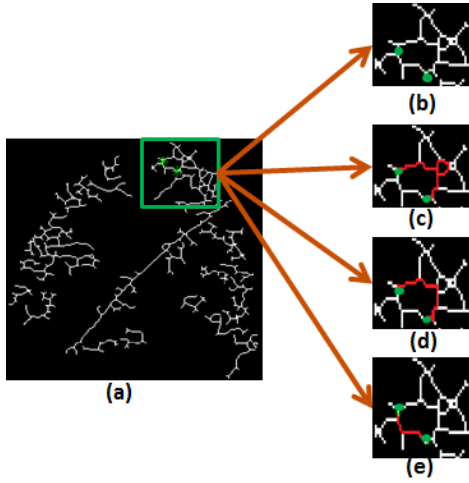


Fig. 3: Example of shortest path finding with image (512×512), (a) skeleton map; (b) data points through which the shortest path is to be found with; (c) path length of 5.9 mm; (d) path length of 5.3 mm; (e) path length of 5 mm (shortest)

$SP_{w,i1,i2}(x,y)$ are determined. This $\mu_{i1,i2}$ and $\sigma_{i1,i2}$ along with the euclidean distance $d_{i1,i2}$ between two centroids c_{i1} and c_{i2} form the basis of our cost function as

$$F(c_{i1}, c_{i2}) = (\mu_{i1,i2} - \sigma_{i1,i2})^2 * d_{i1,i2} \quad (3)$$

Here, $F(c_{i1}, c_{i2})$ is the computed cost function between the centroids c_{i1} and c_{i2} . The underlying assumption for designing such a cost function is based on the fact that the mean intensity of pixels lying on the fissure line will be higher while the variance will be much lower. The second factor ensures the detection of fissure line of longest length. Finally, the line segment $SP_w(x,y)$ in w window segment is selected based on the following equation:

$$\{\hat{c}_{i1}, \hat{c}_{i2}\} = \underset{c_{i1}, c_{i2}}{\operatorname{argmax}} F(c_{i1}, c_{i2}) \quad (4)$$

where, $SP_w(x,y)$ is the optimal path in window w based on centroid $\hat{c}_{i1}, \hat{c}_{i2}$. Thus, the line segment obtained from each window w is merged into a single fissure line image as $F_{line}(x,y) \in \bigcup_w (SP_w(x,y))$. After the exhaustive line fitting over different windows of the image $L(x,y)$, the desired fissure line is selected as the one with the longest length that crosses the median of the lung field area $L(x,y)$ with the end points closer to the lung boundary. The detected fissure line is denoted by $f_{line}(x,y)$.

IV. EXPERIMENTAL ANALYSIS

In this section, the proposed line detection model is validated with a lung analysis dataset and their reference/ground truths.

A. Dataset

A publicly available dataset for lung lobe segmentation algorithms LOLA'11 [16] is used for the evaluation of our proposed method. The dataset contains 55 volumetric chest

CT scans which were acquired from different sites using several different scanners, scanning protocols and reconstruction parameters. Most scans contain various pathological abnormalities ranging from mild to severe. The ground truth of the fissures is manually generated. From the 55 CT scans of publicly available dataset, a subset of 20 cases are selected. The in-plane resolution is between 0.53 mm and 0.78 mm whereas the slice thickness is between 0.3 mm and 1.5 mm.

B. Evaluation

In this section, the proposed fissure line detection technique is validated with the above mentioned dataset and its ground truth. All the implementations are done on MATLAB 2016a version on a 64bit operating system running at 2.81 GHz clock speed. The proposed method is compared with a hessian based filter and fissure detection using line enhancing filter of Klinder et al. [12]. Our proposed method is compared with this paper due to the fact that they have used the same dataset LOLA'11 to validate their result. The proposed method is compared using same 20 subset of the dataset mentioned in the paper [12]. Moreover, the paper has reported a comparison of their result to Hessian filter which is used in the pipeline of the proposed method in a different manner. Klinder et al. [12] have used a line enhancement filter by correlating a model template of the fissure with the observed image data. Please note that we outperform the line enhancement filter [12] even without using any form of template matching and training data.

We have used receiver operating characteristic (ROC) analysis as an evaluation criterion for our method. The ROC curve is a trade-off between two measures, sensitivity and specificity. These measures are based on four conditions namely true positive (TP), false positive (FP), true negative (TN) and false negative (FN). As the pixels of detected fissure may not exactly lie on the ground truth a small tolerance band along the ground truth is used to determine these conditions. The pixels belonging to detected fissure that lie within the tolerance band are considered as TP1 and the rest are FP. Also, the ground truth pixels that corresponds to the detected fissure line are categorized as TP2 and the rest are false negative (FN). The two true positive measures TP1 and TP2 may not be always equal and (TN) are the pixels not belonging to the tolerance band. The sensitivity/recall, specificity and the F-measure are defined as $TP2/(TP2+FN)$, $TN/(TN+FP)$ and $(2 \times \text{Precision} \times \text{Recall})/(\text{Precision} + \text{Recall})$ respectively, where precision is $TP1/(TP1+FP)$. The ROC curve shown in Fig.6 is obtained by varying the tolerance band over all the CT scans. An area under the ROC curve (AUC) is used to measure the performance of the proposed algorithm compared to other methods. An internal comparison has also been performed by setting the second factor ($d_{i1,i2}$) to unity in the cost function (equation 3). This results in performance degradation of AUC by 20% which is also shown in Fig.6. Also, an incomplete fissure line is detected as shown in Fig.7. Table I shows the impact of the factor $d_{i1,i2}$ in the cost function over the performance analysis measures by setting a tolerance band of 3 mm [13]. Setting the factor $d_{i1,i2}$ to unity results a degradation

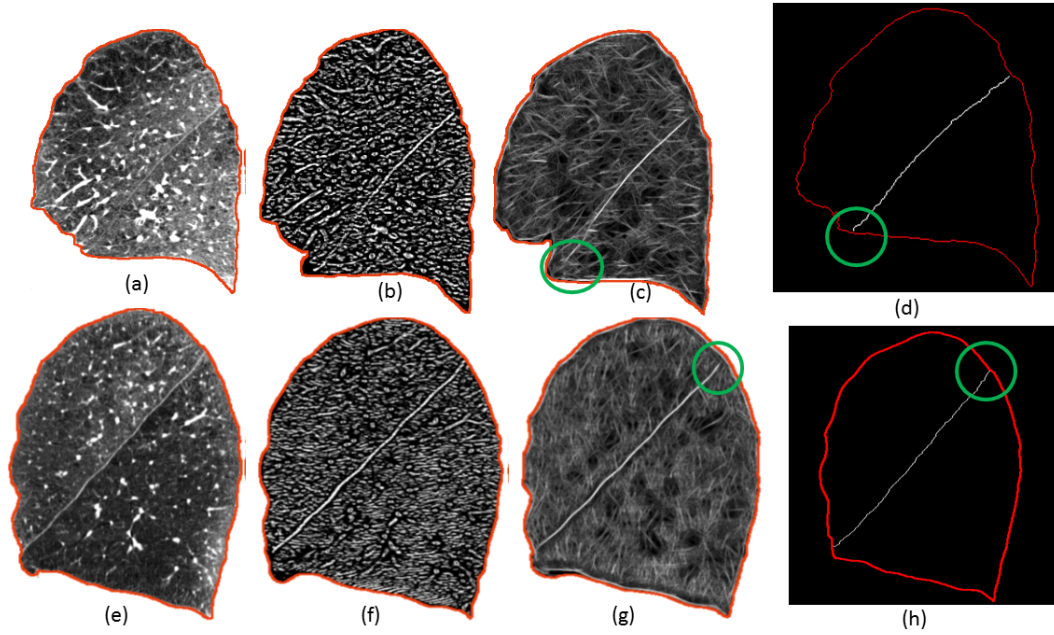


Fig. 4: Comparison of result, (a) Original data; (b) Hessian filter; (c) Line enhancing filter Klinder et al.; (d) Proposed line detection result; (e) Original data; (f) Hessian filter; (g) Line enhancing filter Klinder *et al.*; (h) Proposed line detection result. First row shows case 11 and last row shows case 8.

of sensitivity and F-measure. The specificity does not degrade due to the fact that the number of true negatives (TN) does not change with the absence of the factor $d_{i1,i2}$.

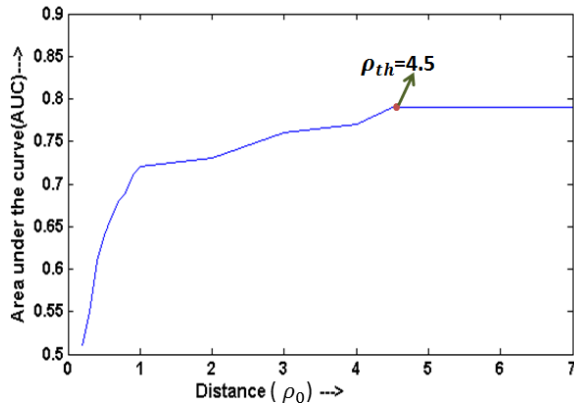


Fig. 5: An example showing the selection of threshold ρ_{th} .

From table II, it is clearly visible that our proposed method

TABLE I: Table Showing the performance analysis over the slices of 20 CT scans

Performance Analysis Measure	Values with $(d_{i1,i2})$	Values without $(d_{i1,i2})$
Average Sensitivity	0.87	0.51
Average Specificity	0.99	0.98
F-measure	0.96	0.62

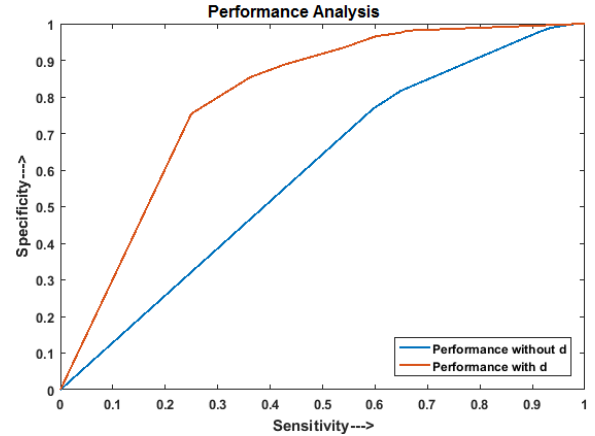


Fig. 6: Performance analysis with and without the second factor $(d_{i1,i2})$ of the cost function (refer equation 3)

TABLE II: Table Showing performance comparison of fissure detection with different methods in terms of AUC

Methods	AUC
Hessian Filter	0.67
Klinder et al.[12]	0.71
Proposed method (with $d_{i1,i2}$)	0.79
Proposed method (without $d_{i1,i2}$)	0.59

has outperformed the other two methods with an area under the curve (AUC) of 0.79 compared to 0.67 using hessian

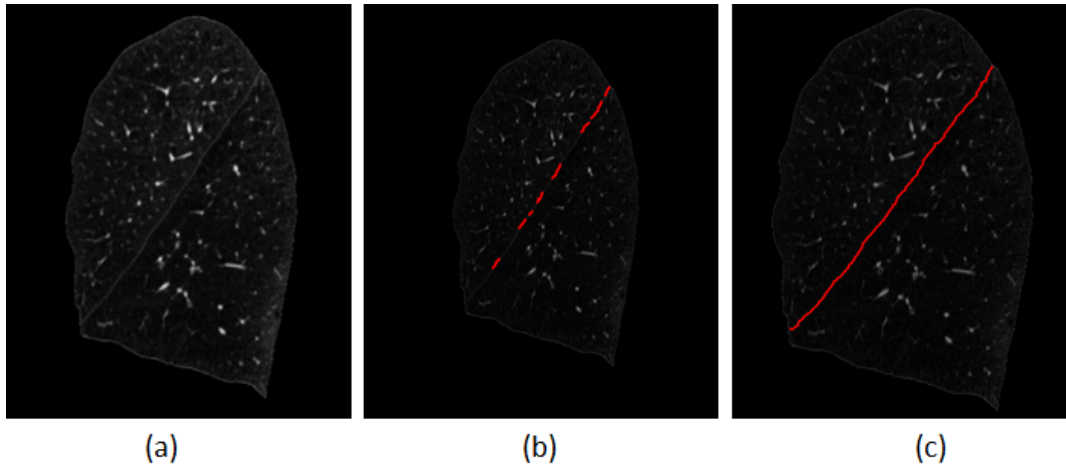


Fig. 7: Visual results with and without the second factor ($d_{i1,i2}$) of cost function (see equation 3), where, (a) an original CT image; (b) incompleteness in fissure detection (without $d_{i1,i2}$); (c) completeness in fissure detection (with $d_{i1,i2}$)

based enhancement filter and 0.71 using line enhancing (LE) filter. We have also included a performance comparison of the proposed method with and without the factor $d_{i1,i2}$ in the cost function. The qualitative result on sagittal view shown in Fig.4 also gives the impression that our proposed method is better than the other two methods. The proposed technique significantly detects the fissure structure while suppressing all other structures while the LE filter and the hessian filter capture other tissues that have similar line like appearance.

V. CONCLUSION

In this paper, we proposed a fissure line detection method using eigen analysis of the Hessian matrix and an exhaustive sample consensus (ESAC) algorithm. The underlying principle is to detect parts of fissure lines within a set of overlapped square windows and then assembling them to obtain the fissure line in its entirety. The fact that the fissure line appears as a piece-wise linear structure in a small window is utilized in our detection scheme. Our line detection technique exhaustively examines all possible combinations of points for accurate detection. A cost function is designed for the selection of the best set of data points through which the fissure line would pass. Overall, we demonstrated detection of fissure line in an unsupervised manner without using any form of template matching and training data at any point in the solution pipeline. In future, the cost function will be modified into a more suitable form considering the scaling issue of mean and variance. Also, the performance of the proposed work will be evaluated on other datasets. We will extend our model to detect 3D fissure surface by integrating information from multiple views.

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